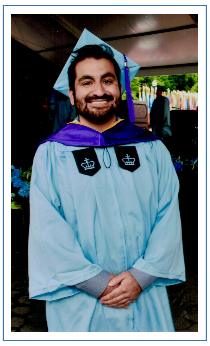

Andrew Quijano

Curriculum Vitae

Andrew Quijano is a current PhD candidate at NYU Tandon, majoring in Computer Science. His dissertation research focuses on creating accurately labeled encrypted network traffic. This would be applicable to creating machine learning models that can make inferences both on human behavior and provide healthcare insights, such as detecting early signs of Alzheimer's or other cognitive impairments.

Education

2023–Present **PhD**, *New York University*, *GPA – 4.0*.
Computer Science, advised by Danny Huang

2021–2022 **Master of Science**, *Columbia University*, *GPA – 3.4*.
Computer Science

2020–2022 **Master of Science**, *New York University*, *GPA – 3.8*.
Cybersecurity

2017–2019 **Bachelor of Science**, *Columbia University*, *GPA – 3.3*.
Computer Science

2015–2017 **Bachelor of Arts**, *The City University of New York*, Queens College, *GPA – 3.5*.
Mathematics

2014–2015 **Associate of Applied Science**, *The City University of New York*, LaGuardia Community College, *GPA – 3.5*.
Computer Operations: Networking and Security

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Andrew's LinkedIn

Experience

Professional

May 2026 – Aug 2026 **PhD SWE Summer Intern**, Google, Kirkland, WA.

- Engineered a deterministic, SQL-driven Perfetto trace evaluation engine to detect Binder IPC transactions and validate AI agent vulnerability report reproduction.
- Benchmarked agent execution across 33 real-world CVEs, achieving 70% automated bug reachability alongside 90% trial-over-trial consistency in both trace reachability and SQL query generation.

Nov 2025 – May 2026 **Senior Information Security Analyst**, Fitch Ratings, New York, NY.

- Optimized organizational risk posture by triaging 100+ high-severity alerts; performed deep-dive analysis on Wiz (CSPM) and Apiiro (ASPM) findings to distinguish between critical production exploits and low-impact package vulnerabilities, significantly streamlining developer remediation efforts.
- Built a Python-based GitHub Actions compliance scanner and dashboard that parsed YAML workflows, detected CI/CD security misconfigurations, and rolled workflow-level findings up to repository-level risk reporting.
- Implemented data validation, error handling, and visualization pipelines using Pandas, Streamlit, and Altair to help engineering and security teams monitor workflow template compliance and prioritize remediation.

Aug 2022 – May 2025 **Application Security Engineer**, Amazon, New York, NY.

- Recognized in **a company-wide email for my security review of Amazon Rufus AI**, where I removed a high-risk administrative permission (SYS_ADMIN_CAP) and prevented legal exposure by halting the use of unvetted third-party AI models and datasets
- Authored internal AWS security standards and developer-centric documentation for S3, Lambda, and KMS; standardized the implementation of automated CloudWatch alerting to detect and mitigate non-programmatic infrastructure changes across service teams
- Remediated 2 critical risks and 80 high-risk vulnerabilities, including **OWASP Top 10** issues like Log4Shell and Credential/PII Logging. This included fixing an outdated TLS vulnerability in more than 3000 internal applications.
- Contributed core modules to BRASR, a cloud-native misconfiguration scanner; unified multiple AWS scanning tools, improved reporting accuracy, and automated false-positive suppression.
- Led security code reviews and validated HIPAA control implementations through **code review and dynamic testing**, ensuring proper encryption, access control, and proper audit logging in native AWS cloud systems.

May 2023 – Aug 2023 **Information Technology Intern**, Hess Corporation, Houston, TX.

- Communicated with vendors to schedule demos and obtain the information required for Hess Security and Operations teams to PoC firewall audit tools
- Completed the design and analysis phase of a Firewall audit tool Proof of Concept
- Presented to IT leadership the business of onboarding a firewall audit tool

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Jul 2019 – Aug 2022 **Security Testing Analyst**, JRI-America, New York, NY.
- Engineered PowerShell and Python automation to manage service accounts, entitlement reviews, and empty AD Group cleanup; eliminated 80% of recurring policy violations.
- Built reporting pipelines to visualize SLA-delinquent vulnerabilities, enabling security posture monitoring at the executive level.
- Developed a password file remediation process in Python that incorporates automated escalation and remediation tracking, resulting in the removal of more than 100 offending files in the shared drives
- Coordinated phased TLS 1.2 upgrades across legacy applications, working cross-functionally with IT, infrastructure, and QA.
- Deployed and tuned InsightAppSec to scan external applications for **OWASP Top 10** vulnerabilities; two XSS flaws were exposed and remediated, improving the security posture with measurable risk reduction.

Feb 2020 – Aug 2020 **Technical Consultant**, Iftach Group, New York, NY.
- Conducted a detailed analysis of potential business partners for MOBI, an Israeli startup focused on using AI for traffic analysis
- Investigated the impacts of the COVID-19 pandemic on MOBI and its potential clients

Oct 16 – May 2017 **Physics Lab Assistant**, CUNY Queens College, Flushing, NY.
- Set up the Physics experiments for Physics Lab sections
- Learned to solder to repair electronics and other equipment in the Physics Lab

Jun 2016 – Aug 2016 **Information Technology Intern**, US Tennis Association, Flushing, NY.
- Assisted in updates and upgrades of loaner laptops for the U.S. Open
- Installed new computers in various offices in the Billie Jean King National Tennis Center
- Created a detailed inventory of all equipment in the IT department and checked if all equipment was functional for the US Open
- In collaboration with other Diversity and Inclusion interns, proposed solutions to the USTA on how to make tennis more popular with people between the ages of 18 and 40

Oct 2015 – Oct 2016 **Bibliographic Researcher**, Micropalentology Press, Flushing, NY.
- Created citation indexes of micro-fossils from Micro-paleontology journals containing the species name and the scientist who discovered it
- Recorded new genus or species discovered or reclassification and corrected on inconsistent species name, plates, or figure numbers

Jun 2015 – Aug 2015 **Information Technology Intern**, LaGuardia Community College, Long Island City, NY.
- Prepared the computer lab for the upcoming Fall 2015 semester by installing new applications, updating software, and creating new partitions
- Responsible for troubleshooting any technical issues with the computers on the LaGuardia network
- Assisted students in configuring/troubleshooting Cisco routers, switches, and firewalls

Teaching Experience

Jan 2023 – May 2024 **Adjunct Professor**, New York University, Brooklyn, NY.
- Created Gradescope autograders (in Python/Bash) for assignments in C, Android, and Django, improving reproducibility and reducing manual grade effort by 80%.
- Designed and graded hands-on security assignments aligned with **OWASP Top 10**, covering topics like SQL injection, Security Logging, and Broken Access Controls

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Sep 2020 – Dec 2021 **Security I Course Assistant**, Columbia University, New York, NY.
- Was a course assistant for the Fall 2020 and Fall 2021 semesters
- Provided documentation on **threat modeling** with an example using an attack tree to analyze the threat of a compromised Columbia UNI credential
- Provided instructions on how to use Travis CI, so students can use CI to confirm that their programs pass their own test cases
- Hosted office hours, answered most course questions offline, and graded homework assignments
- Provided new questions and solutions for the midterm exam and final exam

Jan 2021 – May 2021 **Computer Networks Course Assistant**, Columbia University, New York, NY.
- Held office hours, answered students' questions, and graded homework assignments
- Managed the TA team to ensure timely grading, late-day management, and course logistics
- Provided skeleton code to the class to provide guidance on homework assignments

Jan 2019 – May 2020 **Security II Course Assistant**, Columbia University, New York, NY.
- Was a course assistant for the Spring 2019 and Spring 2020 semesters
- Hosted office hours, answered students' questions, and graded homework assignments
- Provided a lecture on Wireless Security during the Professor's absence

Sep 2018 – May 2019 **Intro to EE Course Assistant**, Columbia University, New York, NY.
- Hosted office hours, answered students' questions, and graded homework assignments
- Updated Homework solutions with more steps to better prepare students for exams

Feb 2017 – May 2017 **Math Lab Teaching Assistant**, CUNY Queens College, Flushing, NY.
- Assisted students in learning Calculus, Discrete Mathematics, Algebra, etc.
- Submitted a solutions manual for a Discrete Mathematics course for internal publication

Research Experience

Sep 2023 – Present **Graduate Research Assistant**, New York University, New York, NY.
- Automated IoT Inspector's key databases (OUI, MMDB, DuckDuckGo) via a new CI/CD pipeline, future-proofing a tool recognized by publications such as the Washington Post.
- Redesign the UI/UX for human-in-the-loop (HITL) labeling of network traffic, a critical component in IoT Inspector's machine learning pipeline for IoT anomaly detection.
- Utilize Angr, a symbolic execution engine, to plant security bugs in more difficult places in the source code that are difficult for fuzzers to explore.
- Port LAVA's codebase to modern toolchains: upgraded LLVM API from v3.3 to v11 and replaced Python 2 with Python 3 for instrumentation scripts.
- Collaborate with ADWISE Lab on research related to applied cryptography on privacy-preserving decision trees and secure drone collision avoidance protocols.

May 2024 – Aug 2024 **Graduate Research Assistant**, MIT Lincoln Laboratory, Lexington, MA.
- Collaborated with MIT LL staff on threat modeling and cyber risk assessment for securing systems.
- Created, tested, and documented a SysML plugin for MagicDraw to assist with cyber risk assessment methodologies.
- Work and findings over the summer led to a conference paper for INCOSE 2026 (to be submitted).

May 2021 – Aug 2021 **Graduate Research Assistant**, MIT Lincoln Laboratory, Lexington, MA.
- Upgraded zero-knowledge performance testbed by implementing horizontal scaling, bastion proxying, and integration tests
- Written a qualitative analysis report on performers converting real-life problems into a novel zero-knowledge representation

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May 2018 – Sep 2018 **Undergraduate Research Assistant**, Columbia University, New York, NY.
- Created an Android app that can passively collect AP, RSSI, and environmental data
- Tested an indoor room-level localization system using machine learning techniques on collected data

Sep 2017 – Jan 2018 **REU Research Assistant**, CUNY Lehman College, Bronx, NY.
- Worked collaboratively with a team to analyze phylogenetic tree structures
- Performed studies on open questions in computational geometry with applications to biological statistics

May 2017 – Aug 2017 **REU Research Assistant**, Florida International University, Miami, FL.
- Created an open-source, documented, and tested implementation in Java of these partially homomorphic cryptographic algorithms: Paillier, DGK, El-Gamal, and Goldwasser-Micali.
- Programmed an Android app to collect battery and computation times on prototype secure indoor localization systems.
- Implemented a fingerprint server using Java MySQL API to build a privacy-preserving indoor localization system.

Awards and Recognition

2025 Sigma Xi - Scientific Research Honor Society
2023 GEM Fellowship - PhD
2021 Columbia University Course Assistant Fellowship
2020 NYU Cyber Fellows Scholarship
2017 Robert M. Lilley Memorial Scholarship
2017 Louis Stokes Alliance for Minority Participation Scholarship

Papers to be submitted

- **Quijano, Andrew**, Huang, Danny, Dolan-Gavitt, Brendan, “Expanding the Search Space of Automated Bug Injection: Breaking Trigger-Vulnerability Locality” *In Progress, aiming for IEEE S&P 2027*.
- **Quijano, Andrew**, Huang, Danny, Dolan-Gavitt, Brendan, “ARMed and Dangerous: Exploring Architecture-Dependent Automated Bug Injection” *In Progress, aiming for ACM CCS 2027*.
- **Quijano, Andrew**, Dombroski, Chris, Huang, Danny, Baek, Hyunwook, “The Perfetto Paradigm: grounding AI agent vulnerability report reproduction via deterministic traces” *In Progress, aiming for ACM CCS 2027*.

Publications

Zhang, A., Quijano, A., Sarathi, T., & Denney, K. (2026). REAP the Rewards: Bridging Risk Assessment and Secure Application Development Practices [Andrew Quijano and Annie Zhang contributed equally and share first authorship]. *Proceedings of the Systems Engineering Conference for the Europe, Middle East, and Africa Sector of INCOSE (EMEASEC 2026)*.

Luedeman, A., Baum, N., Quijano, A., & Akkaya, K. (2024). Privacy-preserving drone navigation through homomorphic encryption for collision avoidance. *2024 IEEE 49th Conference on Local Computer Networks (LCN)*, 1–6. https://doi.org/10.1109/LCN60385.2024.10639770

Quijano, A., Halkidis, S. T., Gallagher, K., Akkaya, K., & Samaras, N. (2024). Enhanced outsourced and secure inference for tall sparse decision trees. *2024 IEEE International Performance,*

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Computing, and Communications Conference (IPCCC), 1–6. https://doi.org/10.1109/IPCCC59868.2024.10850192

Davidov, N., Hernandez, A., Jian, J., McKenna, P., Medlin, K. A., Mojumder, R., Owen, M., Quijano, A., Rodriguez, A., St. John, K., Thai, K., & Uraga, M. (2021). Maximum covering subtrees for phylogenetic networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, *18*(6), 2823–2827. https://doi.org/10.1109/TCBB.2020.3040910

Kobriger, K., Zhang, J., Quijano, A., & Guo, J. (2021). Out of our depth with deep fakes: How the law fails victims of deep fake nonconsensual pornography. *Rich. JL & Tech.*, *28*, 204–253. https://jolt.richmond.edu/files/2021/12/Kobriger-Final-for-Publication.pdf.

Quijano, A., & Akkaya, K. (2019). Server-side fingerprint-based indoor localization using encrypted sorting. *2019 IEEE 16th International Conference on Mobile Ad Hoc and Sensor Systems Workshops (MASSW)*, 53–57. https://doi.org/10.1109/MASSW.2019.00017

Skills

Programming Languages	C/C++, Java, Bash, SQL, Python, PowerShell, Cloud Development Kit (CDK)
Development Applications	Gradle, REST API Development, Docker, Git, GitHub Actions, Kubernetes
Security Applications	Nexpose, InsightAppSec, QRadar, CyberArk, Varonis, IDA Pro, Burp Suite
Soft Skills	Project Management, Technical Writing, Scientific Research, Process Automation

Certifications

May 2021 - May 2029	GIAC Web Application Penetration Test (GWAPT)
Feb 2022 - Feb 2026	GIAC Incident Handling (GCIH)
Jul 2022 - Jun 2026	GIAC Penetration Tester (GPEN)
Jul 2025 - Jul 2029	GIAC Cloud Penetration Tester (GCPN)

Languages and Nationality

English	**Mothertongue**	
Spanish	**Mothertongue**	*Conversationally fluent*
Nationality	**American and Peruvian**	*Dual Citizen*

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